

Conversion of natural language text to SQL queries using generative AI

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ABSTRACT: With the aim of simplifying database interactions, this project introduces an innovative approach known as Semantic Synthesis, which transforms unstructured natural English language input into precise and contextually appropriate SQL queries. In response to the growing demand for user-friendly database access, cutting-edge Generative Artificial Intelligence (Generative AI) techniques are utilized to bridge the semantic gap between user-intended actions and the complexities of structured query language (SQL). The primary objective of this project is to empower individuals without extensive SQL expertise to interact with databases intuitively and extract knowledge from huge databases which help in making data-driven decisions, thereby enhancing usability and accessibility. To derive meaning and context from user-provided textual queries, the approach utilizes a Large Language Model (LLM) deeply integrated with transformers and a user-friendly User Interface (UI). The resulting system excels at generating SQL queries tailored to user-defined tasks and provides a detailed algorithmic breakdown to elucidate the intricacies of query development. This dual-dimensional output ensures transparency throughout the translation process and enhances comprehension of the artificial intelligence-driven conversion from natural language to SQL. This study explores the effectiveness of Semantic Synthesis across various real-world domains, including E-commerce, healthcare, education, and human resources. The project attempts to prove the system's practical utility and potential impact in diverse professional settings by assessing its accuracy, efficiency, and user-friendliness. This project aims to significantly contribute to the democratization of database interaction by adopting a comprehensive strategy that integrates technological innovation with practical usability, effectively adapting to the intricacies of different industries.

Keywords: Database, Structured Query Language (SQL), Generative AI, Artificial Intelligence (AI), Large Language Model (LLM)

1 INTRODUCTION

In the world of modern data-driven environments, it is becoming more and more clear that a smooth and easily navigable database interface is required. This work initiates a novel endeavor by presenting the notion of Semantic Synthesis, an innovative method intended to remove any technical barriers such as technical proficiency or specialized knowledge in database interaction through the conversion of unstructured natural language text into accurate and contextually relevant SQL queries. The core component of this project is the use of sophisticated Generative Artificial Intelligence (AI) methods, which are carefully used

to close the semantic gap between the complex semantics of structured query language (SQL) and the subtle representations of user meaning. The primary objective of this project is to enable people who might not be very proficient with SQL by giving them the means to interact with databases in an easy-to-use manner. In turn, this empowerment changes the user experience, making it more intuitive and user-friendly while also improving accessibility. The process behind this breakthrough is based on the use of transformers and LLM, which are deeply entwined with neural networks and deep learning models that are part of generative AI.

2 MOTIVATION

SQL database interaction can be difficult for many people, especially those who are not in the database management or data science fields. The project’s goal is to lower barriers by offering a natural language interface, which will increase database accessibility for a larger user population. Driven by the goal of enabling non-technical users, the project aims to provide a system that enables people with little to no SQL experience to build queries with ease. It seeks to improve people’s interactions with the utilization of data by streamlining database interaction.

3 LITERATURE SURVEY

In [1], the authors claim that deep learning neural models constitute a challenge to classic generative phrase structure theories of natural language, as they may be applied to software code. They contend that since the same reasoning hold true for programming languages, attaining great performance on random NLP tasks has little bearing on theories of natural language and generative grammar. They suggest a more precise term, “corpus model,” for these models and contend that current neural models fall short in representing pertinent linguistic events. They contend that although artificial neural networks have shown promise in some scenarios, they have not fundamentally changed our understanding of how people think. In [2], in this study, an automated technique for producing user-friendly images—prompt evolution—is presented. It yields notable gains in image quality by fusing the diversity of evolutionary algorithms with the innate capabilities of generative models. Multi-X formulations, single-objective settings, and multi-objective settings can all be used with this technique. It also imposes variation and selection pressure on concealed states, guidance, and outputs, providing a wealth of research opportunities in this nascent discipline. In [3], the study underscores the need of AI literacy in enabling cognitive engagement with generative AI, which unleashes creative possibilities. It focuses on how young students engage with AI and how that engages them in critical thinking, creativity, and problem-solving. The study discovered that high school pupils in Northern Thailand had favorable opinions about AI and regarded it to be highly useful. Important features include aiding skill development, encouraging collaboration, encouraging accessibility, encouraging iteration, and stimulating creative thinking. The report highlights how important it is to incorporate AI instruction into curricula. In [4], in order to help the neural network concentrate more on the relevant portions of the input and less on the irrelevant parts during prediction tasks, the attention mechanism gives each input item a weight. Recent years have seen a major increase in interest in this mechanism, which has prompted more study and applications across a range of industries. In [5], this study presents a new approach to semantic similarity calculation based on multiple influencing elements, which leads to a more accurate result than the original classical algorithm and enhances accuracy and alignment with experts’ expertise. In [6], over the past 60 years, natural language processing, or NLP, has advanced dramatically. For example, gadgets are now able to process voice instructions, and Google’s machine

translation service can translate between numerous languages. This illustrates the capabilities of learning systems and neural networks. Machine-learning frameworks have advanced at a never-before-seen rate, setting records across a range of industries. In natural language challenges, deep neural architecture models perform more efficiently and introduced the idea of “imperfection.” Nonetheless, the majority of research ignores speakers of other languages by using English as the input or output language. Future research ought to take into account utilizing a larger variety of languages and forecast that deep learning models would advance and eventually become the standard in computational linguistics. In [7], In terms of computation and architecture, convolutional neural networks (CNN) have significantly advanced natural language processing (NLP). Word embeddings are introduced by the Word2vec approach, which enhances NLP performance. Continuous Bag-of-Words (CBOW) and Skip-Gram are two famous algorithms used for learning dense representations of words in a lower-dimensional vector space. These methods are extensively employed in many different NLP applications. In [8], a sophisticated and crucial component of artificial intelligence, natural language processing (NLP) enables chatbots to comprehend human input, convert it into a format that can be processed, and reply to it. To comprehend its complexity, one must have a great commitment and determination. An essential component of contemporary AI systems, interview performance evaluation is anticipated to be automatically assessed as AI advances. In [9], the algorithms for supervised, unsupervised, semi-supervised, and reinforcement learning were examined in this paper along with other machine learning techniques and approaches. In the future, it hopes to create a model based on these methodologies, having contributed to their classification.

4 PROPOSED SYSTEM

The project addresses a critical need in the contemporary digital landscape, where databases serve as repositories for an ever-increasing volume of information. Traditional methods of interacting with databases, predominantly reliant on SQL proficiency, pose challenges for non-technical and non-proficient users. The proposed solution aims to democratize database access, providing an inclusive platform for individuals across diverse domains to effortlessly formulate complex queries using natural language. The success of this solution hinges on a meticulously designed set of core mechanisms, each serving a distinct role to advance the overarching objectives of the project.

Comprehending and extracting meaningful information from raw natural language queries is achieved through the implementation of advanced Large Language Models (LLM) and Transformers. These algorithms, encompassing tokenization, part-of-speech tagging, named entity recognition (NER), and lemmatization, collectively form a sophisticated framework for processing and analyzing user input. The objective is clear: enhance the system’s ability to decipher the intricacies of human language, paving the way for more accurate and context-aware query interpretation.

Moving beyond raw input understanding, the project incorporates Intent Classification and Entity Recognition mechanisms. Through the implementation of machine learning classifiers and neural network-based models, the system endeavors to unravel the user’s intent and pinpoint entities crucial for query construction. This layer of sophistication ensures that the generated queries not only capture the literal meaning of the input but also encapsulate the user’s underlying purpose.

The crux of the system lies in the Generative AI Model (Seq2Seq), tasked with translating processed natural language queries into precise SQL queries. Leveraging the Seq2Seq model, fortified with attention mechanisms, this mechanism attains contextual translation, aligning the generated SQL queries with the nuanced intent of the user. This step represents a pivotal bridge between natural language comprehension and SQL query formulation, marking a significant leap in user-friendly database interaction.

Ensuring transparency in the query generation process, the Algorithmic Explanation Generation mechanism steps in. By integrating rule-based algorithms and potentially natural language generation models, the system creates detailed explanations for the generated SQL queries. This not only aids users in understanding the rationale behind the system’s decisions but also contributes to a more user-centric and educational interaction paradigm. In essence, the project orchestrates these core mechanisms seamlessly, weaving together the intricacies of language processing, machine learning, and generative artificial intelligence.

5 ARCHITECTURE

The system architecture of the project seamlessly intertwines Natural Language Processing (NLP) and Deep learning, incorporating sophisticated machine learning and advanced linguistic analysis. The primary modules are designed to synergize their functionalities, fostering a cohesive system that transforms raw natural language queries into precise SQL expressions.

The Natural Language Processing (NLP) Module serves as the foundational layer, tasked with processing and analyzing raw natural language queries. This involves intricate processes such as tokenization, which dissects sentences into individual tokens; part-of-speech tagging, identifying grammatical parts of speech for each token; Named Entity Recognition (NER), detecting entities like table names, attributes, and conditions; and Lemmatization, reducing words to their base or root form. This comprehensive NLP layer forms the basis for the system’s understanding of user input.

Moving forward, the Intent Classification and Entity Recognition Module complements the NLP layer by determining the user’s intent and recognizing pertinent entities. Machine Learning Classifiers, drawing insights from historical data, classify user intent, while Neural Network-based models enhance the system’s ability to recognize entities within the processed natural language. This module ensures a nuanced understanding of user queries, refining the context for subsequent stages.

The Generative AI Model, employing a Seq2Seq architecture, is a pivotal component designed to translate processed natural language queries into SQL expressions accurately. The Sequence-to-Sequence (Seq2Seq) Model, functioning as an encoder-decoder architecture, is augmented by an Attention Mechanism, allowing the model to focus adeptly on relevant segments of the input during the translation process. This integration of advanced AI mechanisms bridges the gap between natural language comprehension and SQL query generation.

The Algorithmic Explanation Generation Module complements the system by providing transparent insights into the process of query generation. Rule-Based Algorithms construct explanations based on predefined rules, ensuring users can comprehend the rationale behind the system’s decisions. There is potential integration of Natural Language Generation Models, introducing coherent text generation for a more user-friendly explanation interface.

The Cross-Domain Adaptability Module further extends the system’s versatility by handling queries from various domains. This module incorporates fine-tuned Domain-Specific Language Models, enabling the system to understand and generate queries specific to different domains. Dynamic Training Data Selection algorithms enhance adaptability by selectively incorporating domain-specific training data, ensuring the system’s flexibility across diverse contexts.

In essence, the overall system architecture seamlessly integrates these modules, creating a sophisticated framework where NLP serves as the initial processing layer, followed by intent classification, SQL query generation, and algorithmic explanation. This cohesive design ensures a comprehensive and user-centric approach to converting natural language queries into SQL expressions, laying the foundation for an accessible and intelligent database interaction system. Figure 1 shows the architecture diagram.

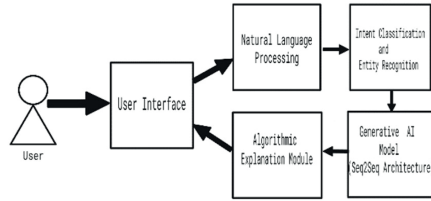


Figure 1. Architecture diagram. ↩

6 CHALLENGES

6.1 *Adaptation to colloquial language*

Users may ask questions in nonstandard or colloquial language. It can be challenging to modify the system such that it can understand and process a large range of linguistic styles.

Interpretability and Explainability: It can be difficult to comprehend how generative models make decisions and to clearly explain the meaning behind the SQL queries that are produced. User trust may be hampered by this lack of interpretability.

6.2 *Error handling*

Systems could find it difficult to accept user mistakes or partial requests with grace. Strong error-handling procedures are needed to provide suggestions or corrections that are significant.

6.3 *Security and privacy issues*

Security issues arise when SQL queries are generated from user input, particularly when the system has strong safeguards against fraudulent queries and unwanted access to private data.

6.4 *Adapting to industry-specific regulations*

There are frequently industry-specific laws governing the use and access to data. Compiling the system with industry-specific requirements adds another layer of intricacy.

6.5 *Education and user onboarding*

Effective education and onboarding tactics are necessary when introducing the system to users who are not familiar with the concept of natural language to SQL conversion. It is crucial to make sure users are aware of the system’s limitations and capabilities.

7 RESULTS

Upon completion of the project “Conversion of Natural Language Text to SQL Queries using Generative AI”, the system is remarkably adept at enabling user-friendly database interaction. Users input queries in natural language, and the system swiftly crafts precise SQL queries based on the schema provided by developers. Moreover, by deconstructing the methodical process of query construction into algorithmic descriptions, the system offers users priceless insights. The dual-layered output not only secures precise translation from natural language to SQL but also amplifies user comprehension and confidence in the system’s capabilities. By seamlessly bridging the semantic gap between user-conveyed intentions and

SQL syntax intricacies, the project successfully democratizes database interaction, empowering individuals across diverse domains to effortlessly extract insights and leverage the power of relational databases with unprecedented ease and efficiency.

Table 1 illustrates the superiority of the proposed system over existing systems in terms of accessibility, approach, algorithmic description, and adaptability to domains. It demonstrates how the proposed system outperforms its counterparts by offering a more user-friendly interface, a different approach, comprehensive algorithmic descriptions, and the ability to seamlessly adapt to various domains with ease.

Table 1. Comparison between proposed system and existing systems. ↩

Aspect	This project	Existing systems
Accessibility	Enhances accessibility by offering a natural language interface, empowering users with limited proficiency to interact with databases.	Accessibility features may vary, with some systems lacking intuitive interfaces or support for non-technical users.
Approach	Utilizes Generative AI for generating SQL queries when natural language input is given.	Various approaches, including rule-statistical, and machine learning methods.
Algorithmic Description	Provides algorithmic breakdowns for transparency and understanding.	May lack detailed explanations or provide non-specific and limited insights into the query generation.
Adaptability to Domains	Demonstrates adaptability across various domains and industries.	May have limited adaptability to specific domains, requiring extensive customization.

Figure 2 shows the user interface of the working project. The left side of the UI is for entering schema, the right side is for entering user prompts, generated queries, and algorithmic description. Figure 3 shows a schema that is to be entered by the developers. Figure 4 shows that on entering user prompts, SQL query is generated and the algorithmic description for a particular query is generated.

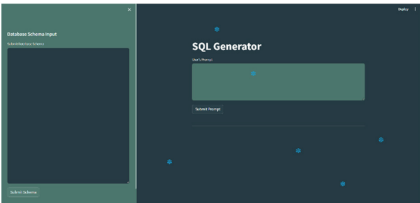


Figure 2. User interface of the working model. ↩

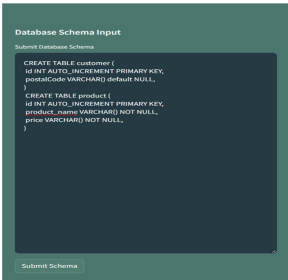


Figure 3. Schema input. ↩

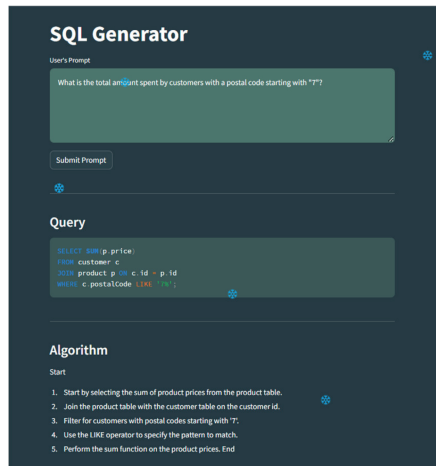


Figure 4. SQL query generation. ↗

8 FUTURE WORKS

8.1 *Multi-language support*

Future works might concentrate on making the system multilingual, accommodating a wider range of users, and supporting international applications.

8.2 *Real-time collaboration and query refinement*

Real-time collaboration features can be enabled, allowing numerous users to work together to formulate complex queries, with the system providing immediate feedback and suggestions for query refinement.

8.3 *User confidence metrics*

Creating measures and techniques to gauge user confidence in the SQL queries that are generated, allowing the system to deliver more precise and contextualized results according to the user's confidence level.

8.4 *Ethical considerations and bias mitigation*

Research and development is necessary to handle moral issues, lessen prejudices, and guarantee that the system reacts fairly to a variety of user types and usage circumstances.

8.5 *Continuous learning and dynamic adaptation*

Implementing a continuous learning mechanism that allows the system to dynamically adapt and evolve over time based on user interactions, feedback, and changes in natural language patterns.

9 CONCLUSION

In conclusion, the project "Conversion of Natural Language Text to SQL Queries using Generative AI" represents a significant advancement in the field of database interaction and

natural language processing (NLP). Through the integration of advanced generative AI techniques and deep learning models, the system successfully bridges the semantic gap between user-conveyed intentions and the intricacies of structured query language (SQL). The project achieves its core objective of democratizing database interaction by empowering individuals with limited SQL proficiency to seamlessly formulate queries in natural language, thereby enhancing accessibility and usability across diverse domains and industries.

The construction of end-to-end applications using Generative AI, exemplified by the “Conversion of Natural Language to SQL Queries” project, holds immense promise in revolutionizing the landscape of human-computer interaction. The results obtained underscore the significance and potential impact of such applications. The precision achieved in natural language processing, coupled with accurate intent classification and entity recognition, demonstrates the system’s adeptness in understanding and interpreting user queries.

The performance of the Generative AI Model, particularly its ability to translate natural language queries into SQL expressions, marks a significant advancement in making complex database interactions accessible to a broader audience. Additionally, the provision of detailed algorithmic breakdowns enhances transparency and fosters user understanding of the query generation process, instilling confidence in the system’s capabilities.

The results obtained from this project lay the groundwork for future advancements in end-to-end applications utilizing Generative AI. As we look ahead, the integration of such technologies opens doors to new possibilities, empowering users with sophisticated tools that bridge the gap between human language and machine understanding. This research paper serves as a testament to the potential of Generative AI in constructing intelligent, user-centric applications, setting the stage for a transformative era in human-computer interaction.

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